

Lab Experiments

Date:17-07-26

Questions:1-5

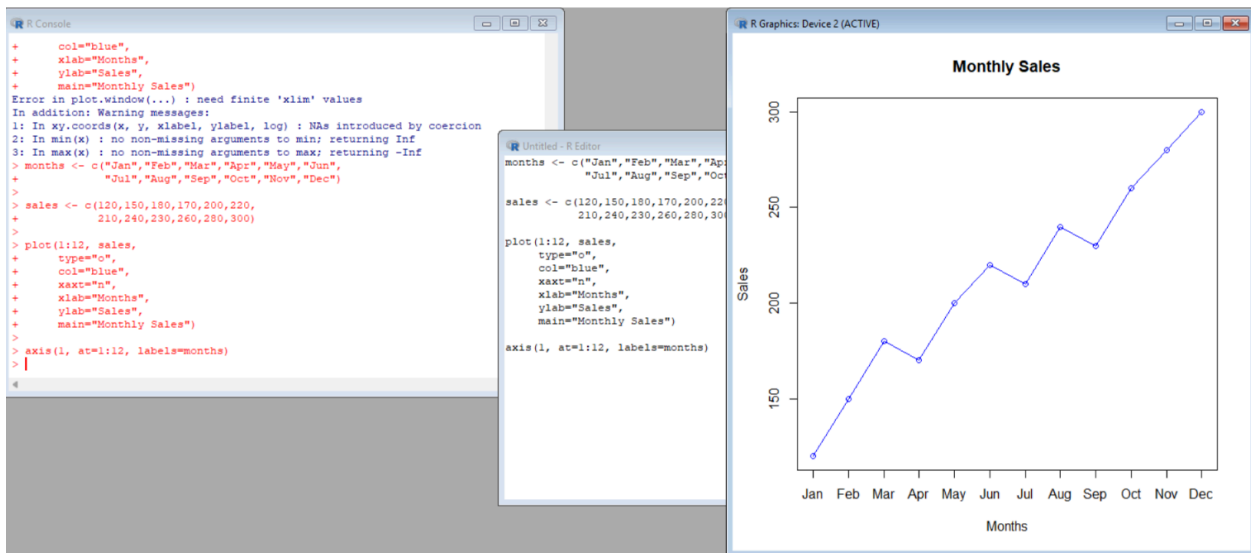
Set 1: Monthly Sales Data

Q1. Line Chart – Monthly Sales

Code:

```
months <- c("Jan","Feb","Mar","Apr","May","Jun",
            "Jul","Aug","Sep","Oct","Nov","Dec")
sales <- c(120,150,180,170,200,220,
          210,240,230,260,280,300)
plot(1:12, sales,
     type="o",
     col="blue",
     xaxt="n",
     xlab="Months",
     ylab="Sales",
     main="Monthly Sales")
axis(1, at=1:12, labels=months)
```

Output:

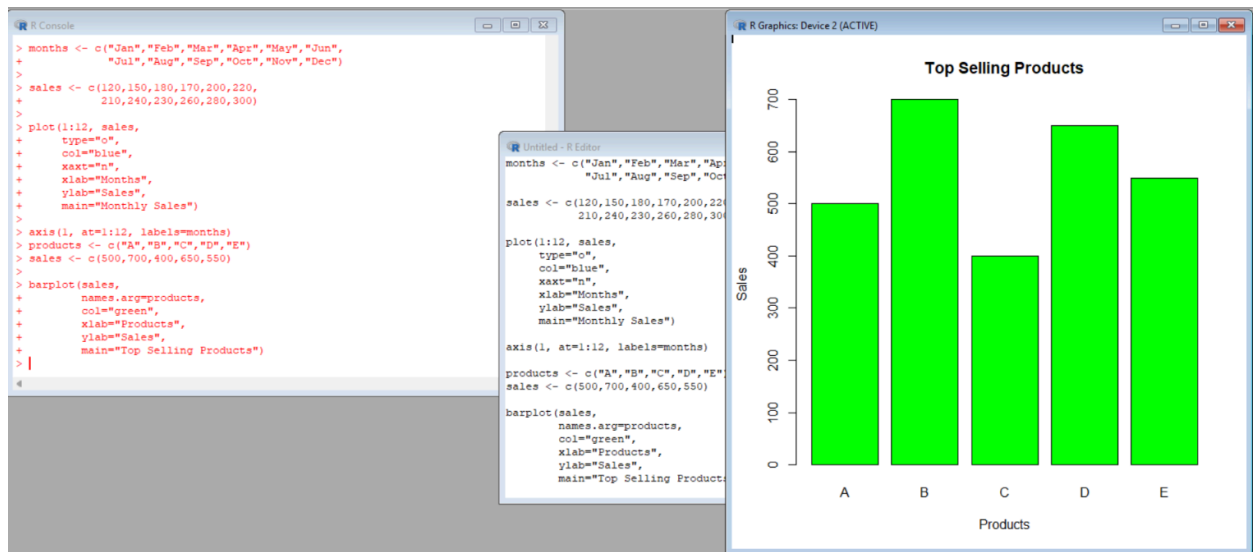


Q2. Bar Chart – Top Selling Products

Code:

```
products <- c("A","B","C","D","E")
sales <- c(500,700,400,650,550)
barplot(sales,
  names.arg=products,
  col="green",
  xlab="Products",
  ylab="Sales",
  main="Top Selling Products")
```

Output:

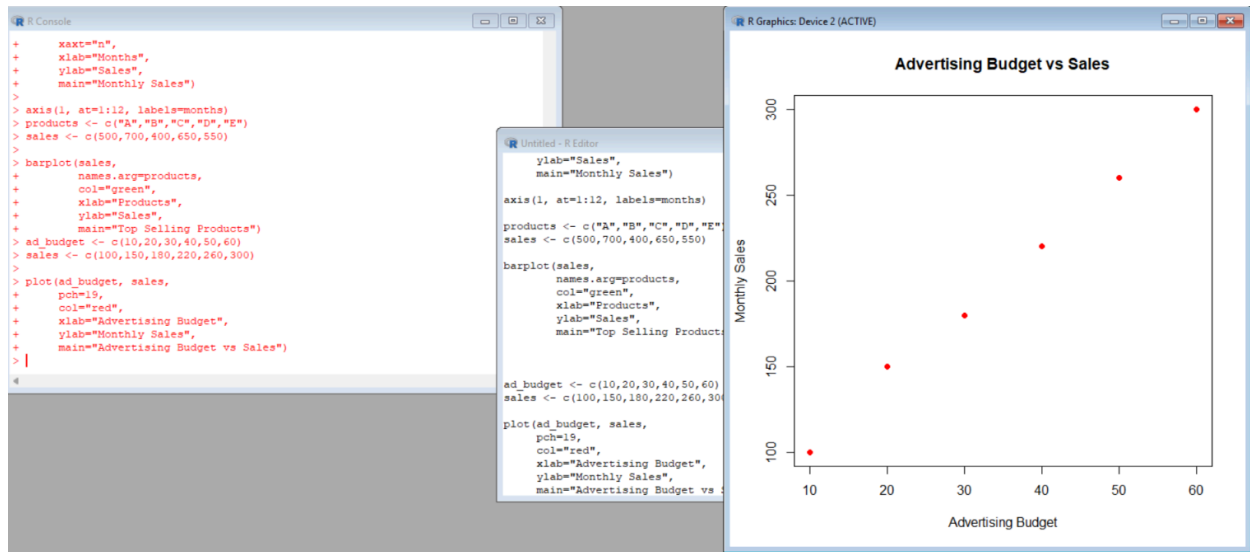


Q3. Scatter Plot – Advertising Budget vs Monthly Sales

Code:

```
ad_budget <- c(10,20,30,40,50,60)
sales <- c(100,150,180,220,260,300)
plot(ad_budget, sales,
  pch=19,
  col="red",
  xlab="Advertising Budget",
  ylab="Monthly Sales",
  main="Advertising Budget vs Sales")
```

Output:



Q4. Interactive Dashboard (Shiny)

Code:

```

library(shiny)
ui <- fluidPage(
  titlePanel("Sales Dashboard"),
  plotOutput("line"),
  plotOutput("bar")
)
server <- function(input, output) {
  output$line <- renderPlot({
    months <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
               "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
    sales <- c(120,150,180,170,200,220,
               210,240,230,260,280,300)
    plot(1:12, sales,
         type="o",
         xaxt="n",
         col="blue",
         xlab="Months",
         ylab="Sales",
         main="Monthly Sales")
    axis(1, at=1:12, labels=months)
  })
  output$bar <- renderPlot({

```

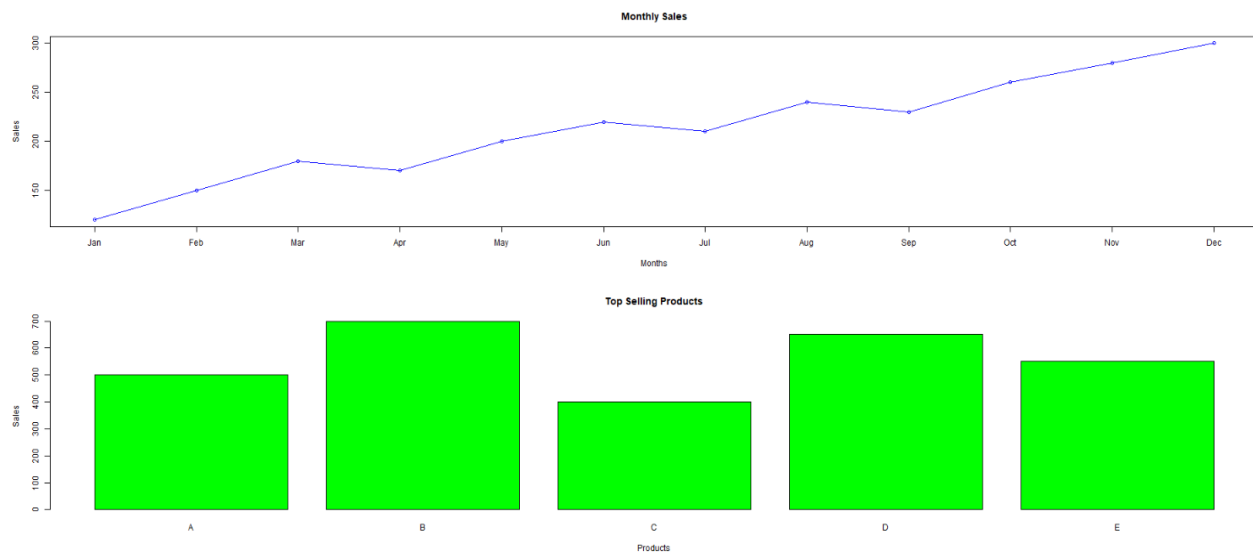
```

products <- c("A","B","C","D","E")
sales <- c(500,700,400,650,550)
barplot(sales,
        names.arg=products,
        col="green",
        xlab="Products",
        ylab="Sales",
        main="Top Selling Products")
})
}
shinyApp(ui, server)

```

Output:

Sales Dashboard



Set 2: Customer Feedback Analysis.

1. Histogram – Distribution of Customer Ages

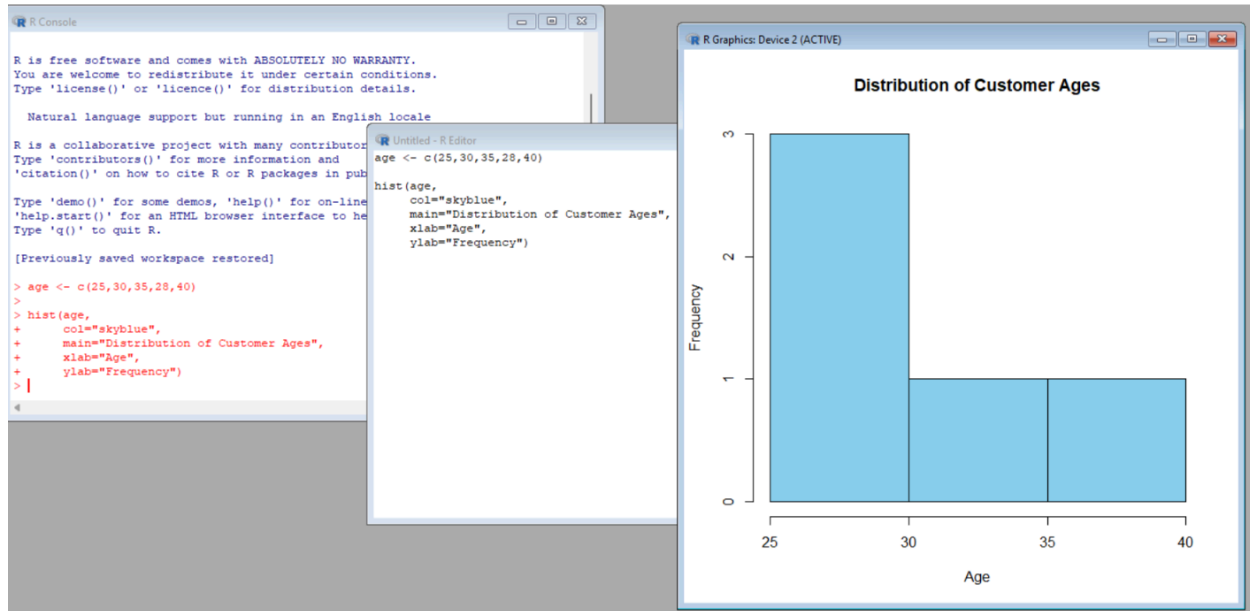
Code:

```

age <- c(25,30,35,28,40)
hist(age,
      col="skyblue",
      main="Distribution of Customer Ages",
      xlab="Age",
      ylab="Frequency")

```

Output:

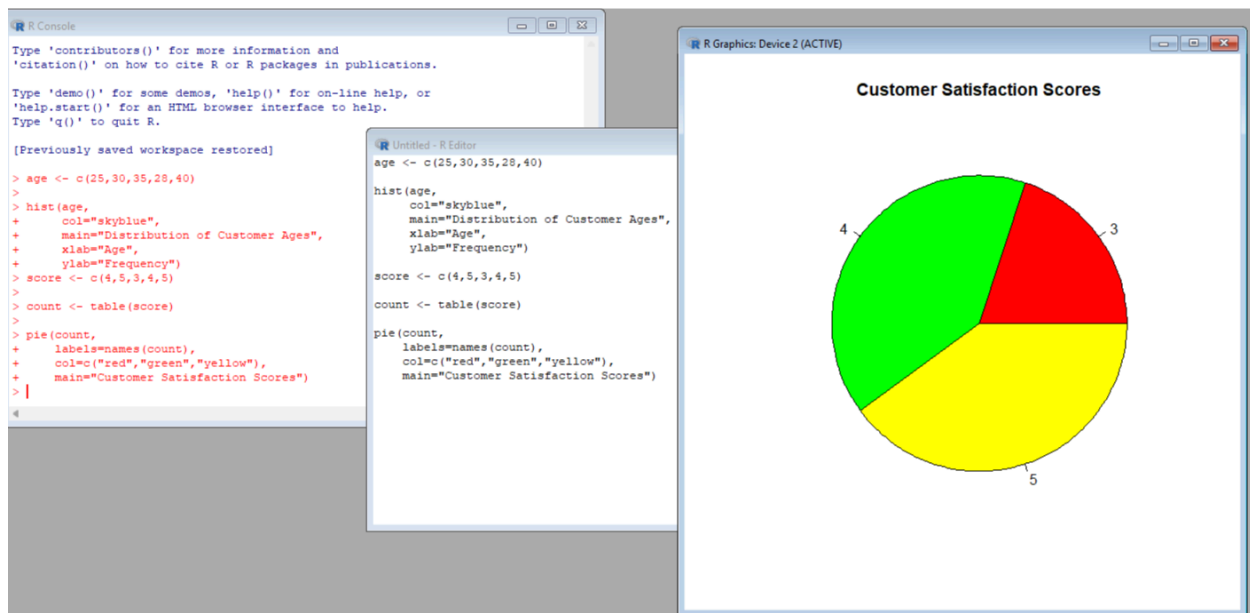


2. Pie Chart – Customer Satisfaction Scores

Code:

```
score <- c(4,5,3,4,5)
count <- table(score)
pie(count,
     labels=names(count),
     col=c("red","green","yellow"),
     main="Customer Satisfaction Scores")
```

Output:

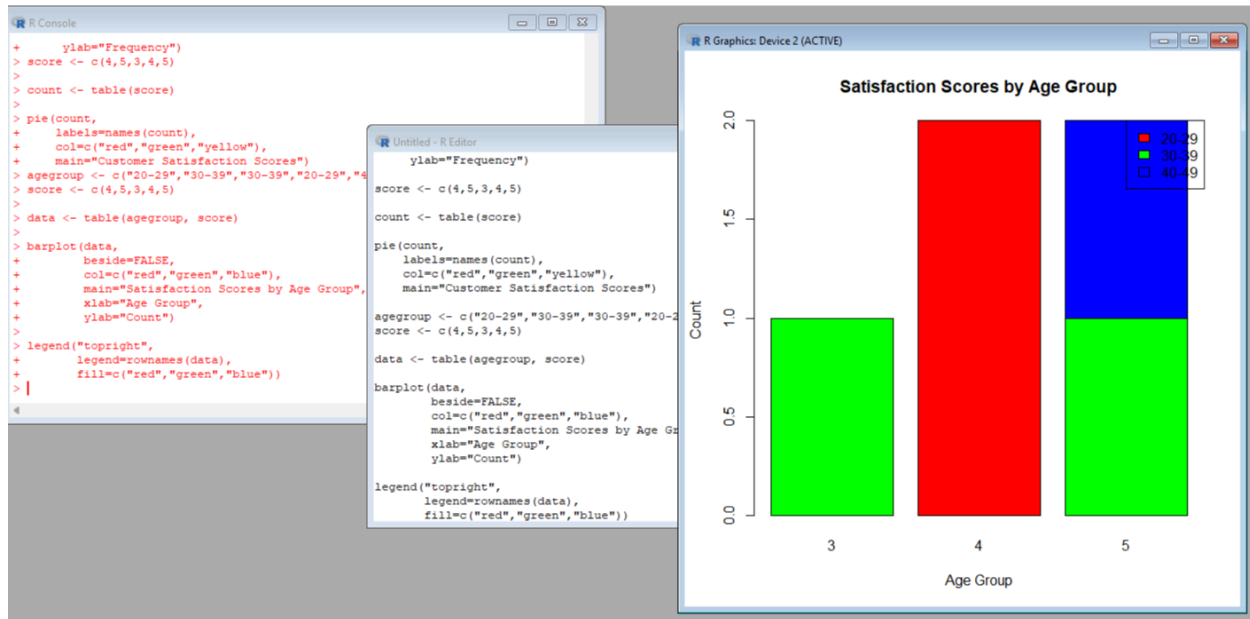


3. Stacked Bar Chart – Satisfaction Scores by Age Group

Code:

```
agegroup <- c("20-29","30-39","30-39","20-29","40-49")
score <- c(4,5,3,4,5)
data <- table(agegroup, score)
barplot(data,
  beside=FALSE,
  col=c("red","green","blue"),
  main="Satisfaction Scores by Age Group",
  xlab="Age Group",
  ylab="Count")
legend("topright",
  legend=rownames(data),
  fill=c("red","green","blue"))
```

Output:



4. Word Cloud – Customer Feedback

Code:

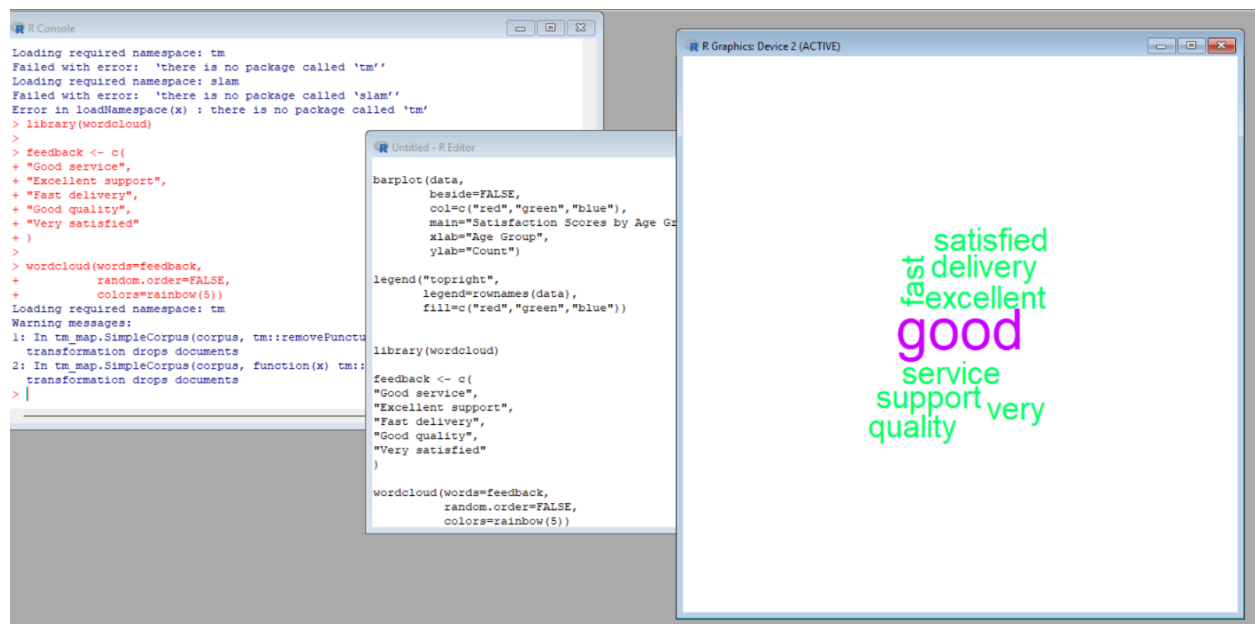
```
library(wordcloud)
feedback <- c(
  "Good service",
  "Excellent support",
```

```

"Fast delivery",
"Good quality",
"Very satisfied"
)
wordcloud(words=feedback,
          random.order=FALSE,
          colors=rainbow(5))

```

Output:



Set 3: Employee Performance Evaluation

Q1. Line Chart – Employee Performance Trend

Code:

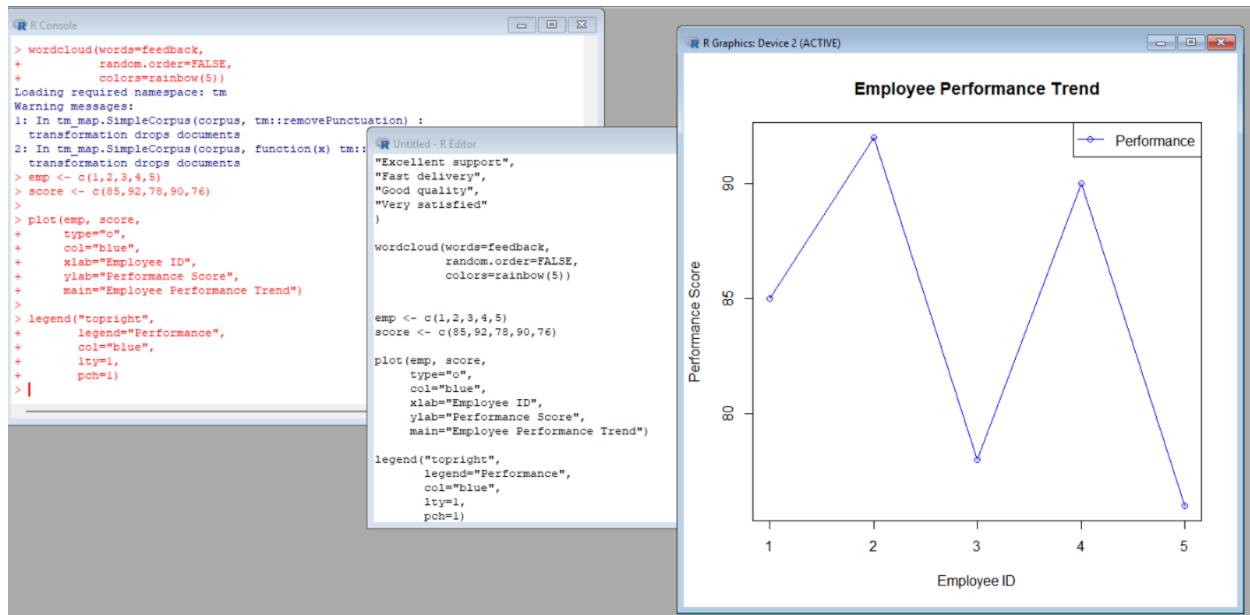
```

emp <- c(1,2,3,4,5)
score <- c(85,92,78,90,76)
plot(emp, score,
     type="o",
     col="blue",
     xlab="Employee ID",
     ylab="Performance Score",
     main="Employee Performance Trend")
legend("topright",
     legend="Performance",
     col="blue",

```

```
lty=1,  
pch=1)
```

Output:

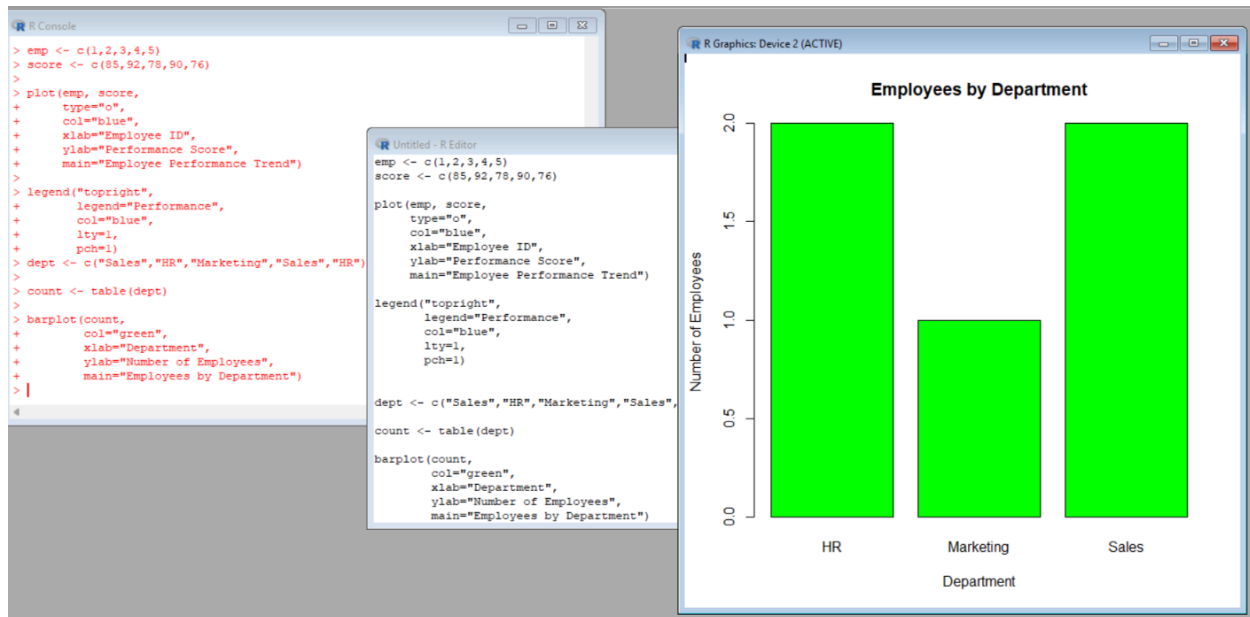


Q2. Bar Chart – Employees by Department

Code:

```
dept <- c("Sales", "HR", "Marketing", "Sales", "HR")  
count <- table(dept)  
barplot(count,  
        col="green",  
        xlab="Department",  
        ylab="Number of Employees",  
        main="Employees by Department")
```

Output:

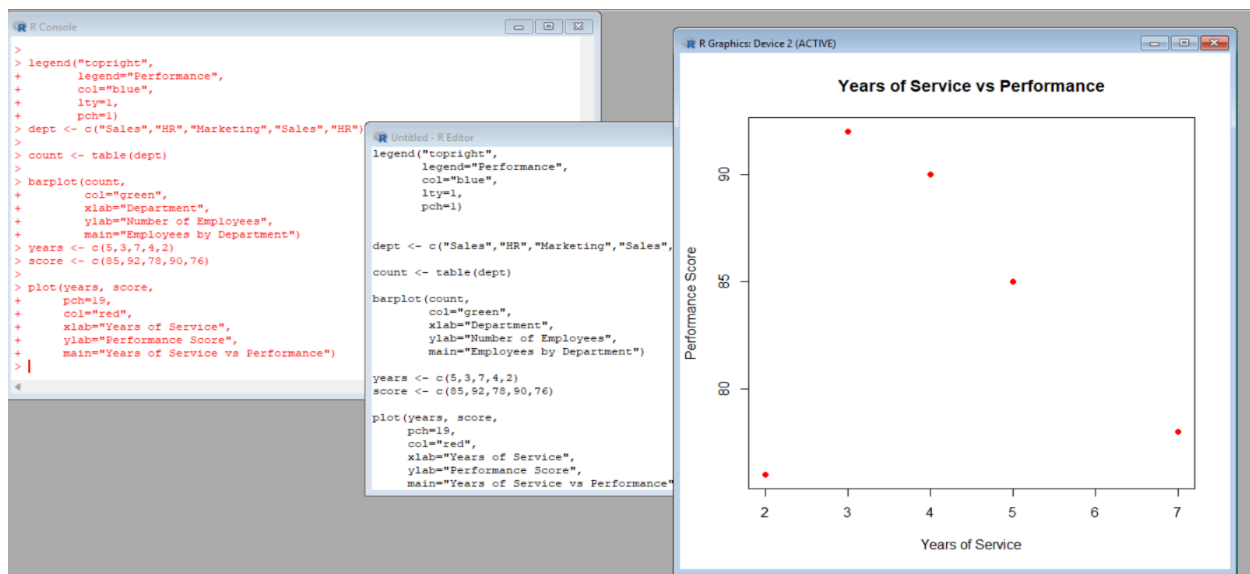


Q3. Scatter Plot – Years of Service vs Performance Score

Code:

```
years <- c(5,3,7,4,2)
score <- c(85,92,78,90,76)
plot(years, score,
      pch=19,
      col="red",
      xlab="Years of Service",
      ylab="Performance Score",
      main="Years of Service vs Performance")
```

Output:



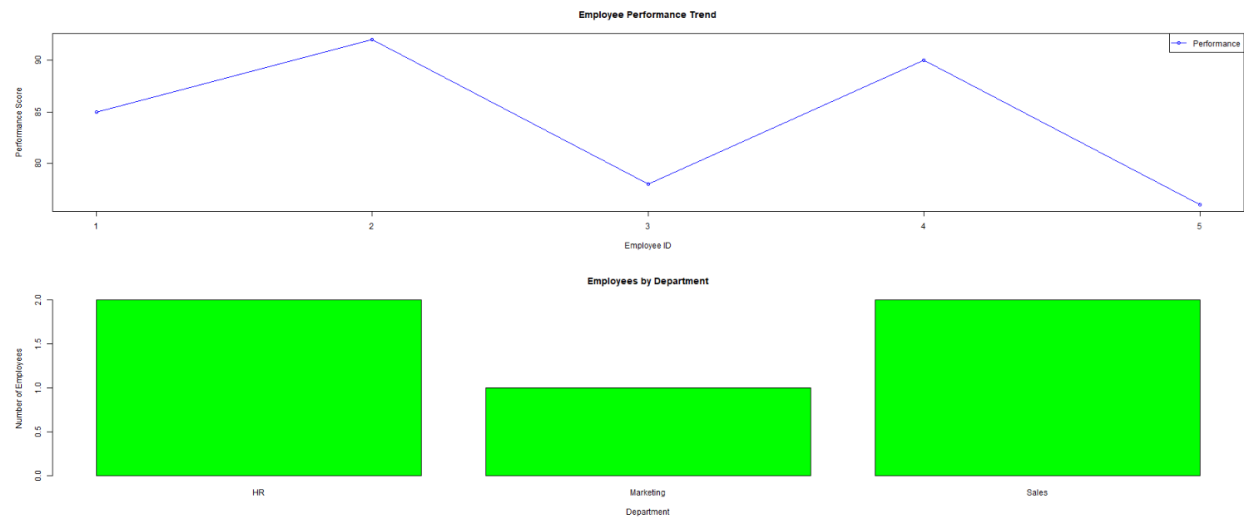
Q4. Interactive Dashboard (Shiny)

Code:

```
library(shiny)
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  plotOutput("line"),
  plotOutput("bar")
)
server <- function(input, output){
  output$line <- renderPlot({
    emp <- c(1,2,3,4,5)
    score <- c(85,92,78,90,76)
    plot(emp, score,
         type="o",
         col="blue",
         xlab="Employee ID",
         ylab="Performance Score",
         main="Employee Performance Trend")
    legend("topright",
          legend="Performance",
          col="blue",
          lty=1,
          pch=1)
  })
  output$bar <- renderPlot({
    dept <- c("Sales","HR","Marketing","Sales","HR")
    count <- table(dept)
    barplot(count,
            col="green",
            xlab="Department",
            ylab="Number of Employees",
            main="Employees by Department")
  })
}
shinyApp(ui, server)
```

Output:

Employee Performance Dashboard



Set 4: Product Inventory Management

Q1. Bar Chart – Quantity of Each Product

Code:

```
product <- c("Product A","Product B","Product C","Product D","Product E")
```

```
quantity <- c(250,175,300,200,220)
```

```
barplot(quantity,  
  names.arg=product,  
  col="skyblue",  
  xlab="Products",  
  ylab="Quantity Available",  
  main="Product Inventory")
```

Output:

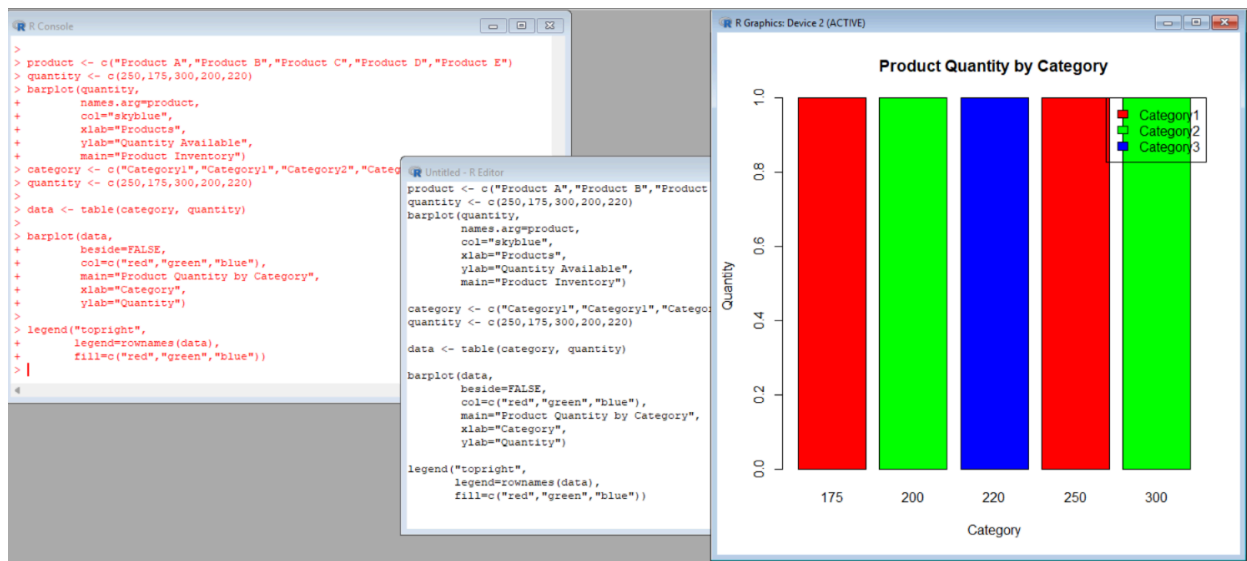


Q2. Stacked Bar Chart – Product Categories

Code:

```
category <- c("Category1","Category1","Category2","Category2","Category3")
quantity <- c(250,175,300,200,220)
data <- table(category, quantity)
barplot(data,
  beside=FALSE,
  col=c("red","green","blue"),
  main="Product Quantity by Category",
  xlab="Category",
  ylab="Quantity")
legend("topright",
  legend=rownames(data),
  fill=c("red","green","blue"))
```

Output:

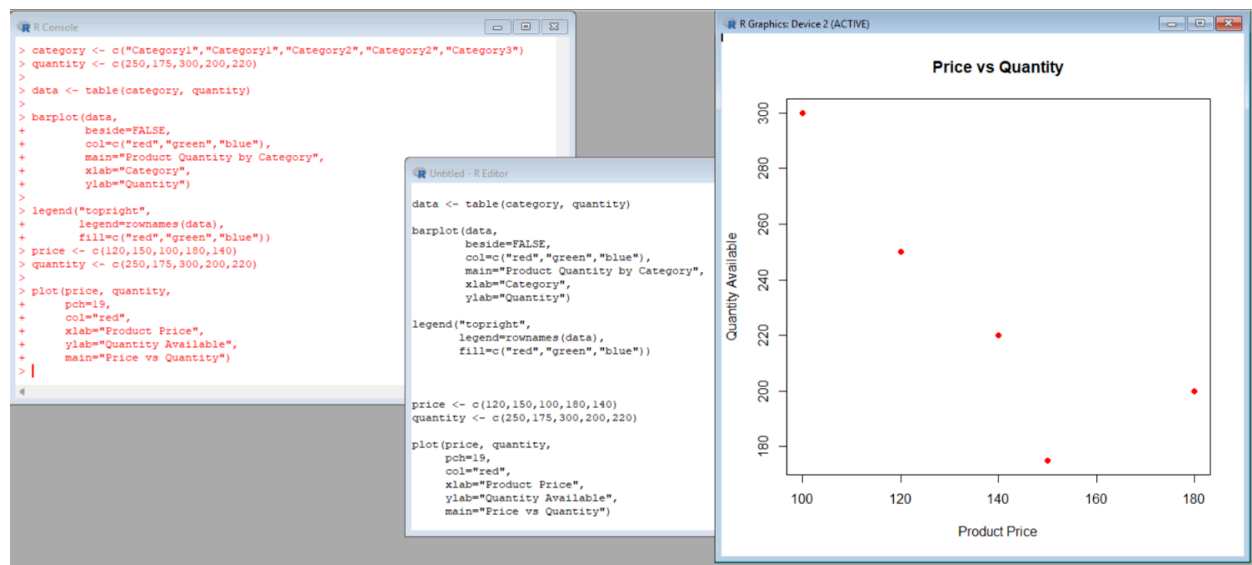


Q3. Scatter Plot – Product Price vs Quantity

Code:

```
price <- c(120,150,100,180,140)
quantity <- c(250,175,300,200,220)
plot(price, quantity,
  pch=19,
  col="red",
  xlab="Product Price",
  ylab="Quantity Available",
  main="Price vs Quantity")
```

Output:



Q4. Interactive Dashboard (Shiny)

Code:

```
library(shiny)
ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  plotOutput("bar"),
  plotOutput("stack")
)
server <- function(input, output){
  output$bar <- renderPlot({
    product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
    quantity <- c(250, 175, 300, 200, 220)
    barplot(quantity,
      names.arg=product,
      col="skyblue",
      xlab="Products",
      ylab="Quantity",
      main="Product Inventory")
  })
  output$stack <- renderPlot({
    category <- c("Category1", "Category1", "Category2", "Category2", "Category3")
    quantity <- c(250, 175, 300, 200, 220)
    data <- table(category, quantity)
    barplot(data,
```

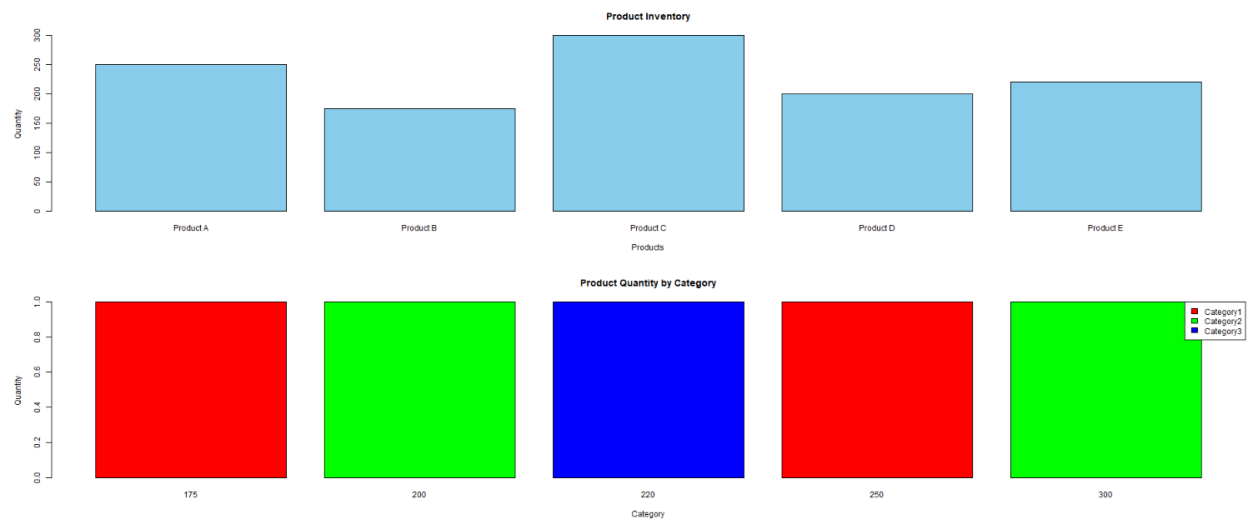
```

        beside=FALSE,
        col=c("red","green","blue"),
        main="Product Quantity by Category",
        xlab="Category",
        ylab="Quantity")
    legend("topright",
          legend=rownames(data),
          fill=c("red","green","blue"))
  })
}
shinyApp(ui, server)

```

Output:

Product Inventory Dashboard



Set 5: Website Analytics

Q1. Line Chart – Daily Page Views

Code:

```

date <- c("Jan1","Jan2","Jan3","Jan4","Jan5")
views <- c(1500,1600,1400,1650,1800)
plot(1:5, views,
     type="o",
     col="blue",
     xaxt="n",
     xlab="Date",
     ylab="Page Views",

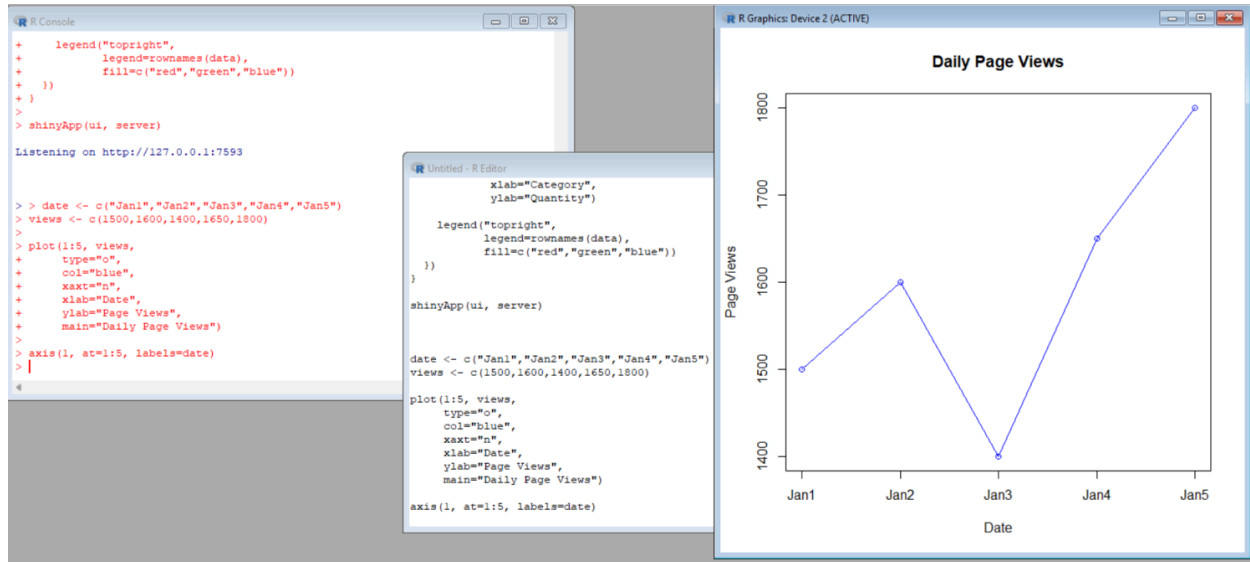
```

```

    main="Daily Page Views")
axis(1, at=1:5, labels=date)

```

Output:



Q2. Bar Chart – Click-through Rate

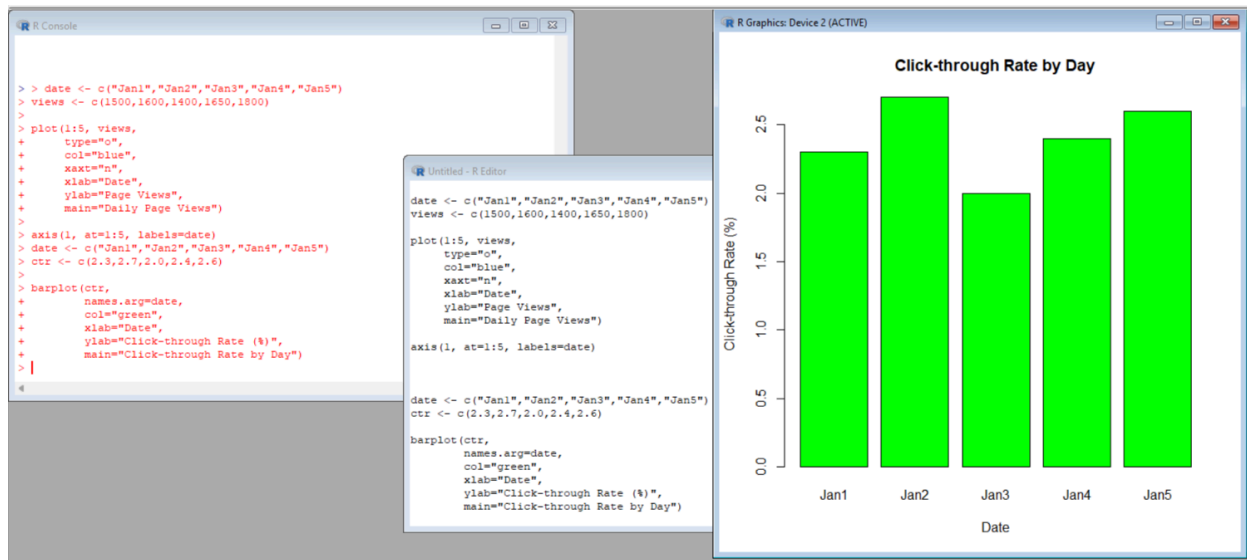
Code:

```

date <- c("Jan1","Jan2","Jan3","Jan4","Jan5")
ctr <- c(2.3,2.7,2.0,2.4,2.6)
barplot(ctr,
        names.arg=date,
        col="green",
        xlab="Date",
        ylab="Click-through Rate (%)",
        main="Click-through Rate by Day")

```

Output:

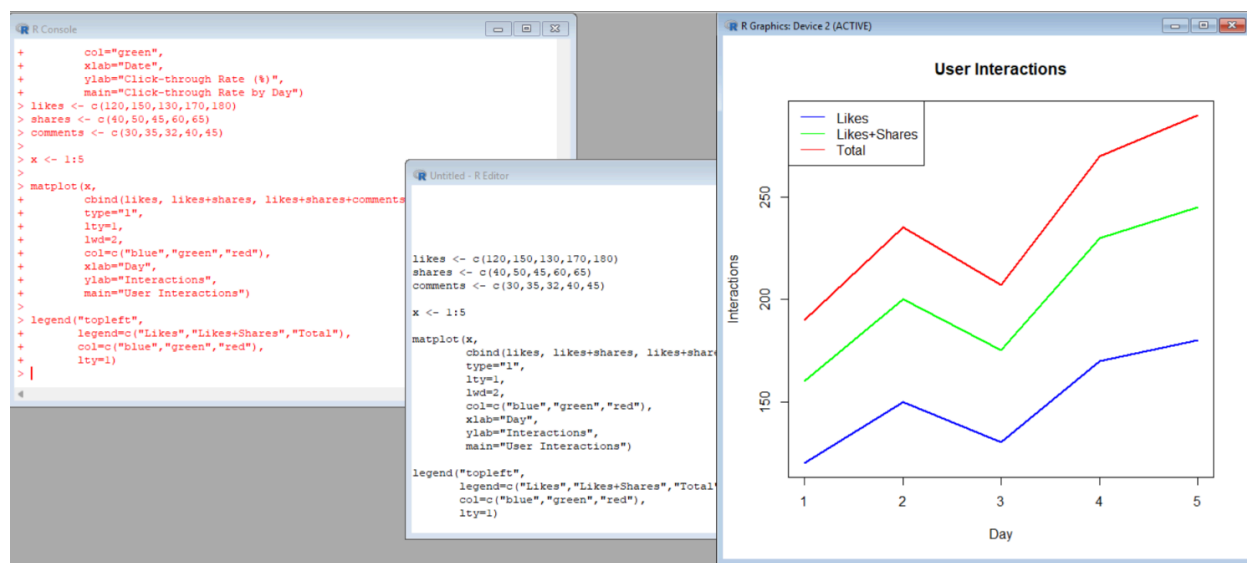


Q3. Stacked Area Chart – User Interactions

Code:

```
likes <- c(120,150,130,170,180)
shares <- c(40,50,45,60,65)
comments <- c(30,35,32,40,45)
x <- 1:5
matplot(x,
        cbind(likes, likes+shares, likes+shares+comments),
        type="l",
        lty=1,
        lwd=2,
        col=c("blue","green","red"),
        xlab="Day",
        ylab="Interactions",
        main="User Interactions")
legend("topleft",
      legend=c("Likes","Likes+Shares","Total"),
      col=c("blue","green","red"),
      lty=1)
```

Output:



Q4. Interactive Dashboard (Shiny)

Code:

```

library(shiny)
ui <- fluidPage(
  titlePanel("Website Analytics Dashboard"),
  plotOutput("line"),
  plotOutput("bar")
)
server <- function(input, output){
  output$line <- renderPlot({
    date <- c("Jan1","Jan2","Jan3","Jan4","Jan5")
    views <- c(1500,1600,1400,1650,1800)
    plot(1:5, views,
         type="o",
         col="blue",
         xaxt="n",
         xlab="Date",
         ylab="Page Views",
         main="Daily Page Views")
    axis(1, at=1:5, labels=date)
  })
  output$bar <- renderPlot({
    date <- c("Jan1","Jan2","Jan3","Jan4","Jan5")
    ctr <- c(2.3,2.7,2.0,2.4,2.6)
    barplot(ctr,

```

```

names.arg=date,
col="green",
xlab="Date",
ylab="CTR (%)",
main="Click-through Rate")
})
}
shinyApp(ui, server)

```

Output:

Website Analytics Dashboard

